```
"""Q1. Percentile using Python
Write a Python program to calculate the 25th percentile, 50th percentile, and 90th percentile
for the following dataset.
Your program should:
• Import NumPy.
• Calculate the 25th, 50th, and 90th percentiles.
• Print all three percentile values.
marks = [42, 48, 51, 55, 58, 60, 63, 67, 70, 72,
75, 78, 80, 83, 85, 88, 90, 92, 95, 98]
"""
#ANS:

#Import NumPy
import numpy as np

marks = [42, 48, 51, 55, 58, 60, 63, 67, 70, 72,
         75, 78, 80, 83, 85, 88, 90, 92, 95, 98]

#Calculate percentiles
p25 = np.percentile(marks, 25)
p50 = np.percentile(marks, 50)
p90 = np.percentile(marks, 90)

# Display results
print("25th Percentile:", p25)
print("50th Percentile (Median):", p50)
print("90th Percentile:", p90)
```

```
25th Percentile: 59.5
50th Percentile (Median): 73.5
90th Percentile: 92.30000000000001
```

```
"""Q2. Quantile using Python
Write a Python program to calculate the quantiles and Interquartile Range (IQR) for the
following dataset.
Your program should:
• Calculate Q1 (25%).
• Calculate Median (Q2).
• Calculate Q3 (75%).
• Calculate the Interquartile Range (IQR).
• Print all values.
sales = [15, 18, 20, 22, 25, 28, 30, 32,
35, 38, 40, 42, 45, 48, 50, 52]
"""
#ANS

Q1 = 24.25
median = 33.5
Q3 = 42.75

IQR = Q3 - Q1

lower_limit = Q1 - (1.5 * IQR)
upper_limit = Q3 + (1.5 * IQR)

print("Q1 =", Q1)
print("Median (Q2) =", median)
print("Q3 =", Q3)
print("IQR =", IQR)
print("Lower Limit =", lower_limit)
print("Upper Limit =", upper_limit)
```

```
Q1 = 24.25
Median (Q2) = 33.5
```

```
Q3 = 42.75
IQR = 18.5
Lower Limit = -3.5
Upper Limit = 70.5
```

```
"""Q3. Boxplot using Python
Write a Python program to create a boxplot for the following dataset and identify any outliers.
Your program should:
• Draw a boxplot using Matplotlib.
• Calculate Q1, Q3, and IQR.
• Calculate the lower fence and upper fence.
• Print all outliers in the dataset.
visitors = [-1 ,-30,22, 25, 27, 30, 32, 35, 36, 38, 40,
42, 45, 48, 50, 52, 55, 58, 90, 95]
"""
#ANS:

import numpy as np
import matplotlib.pyplot as plt

visitors = [-1, -30, 22, 25, 27, 30, 32, 35, 36, 38,
            40, 42, 45, 48, 50, 52, 55, 58, 90, 95]

Q1 = np.quantile(visitors, 0.25)
median = np.quantile(visitors, 0.50)
Q3 = np.quantile(visitors, 0.75)

IQR = Q3 - Q1

lower_limit = Q1 - (1.5 * IQR)
upper_limit = Q3 + (1.5 * IQR)

print("Q1 =", Q1)
print("Median (Q2) =", median)
print("Q3 =", Q3)
print("IQR =", IQR)
print("Lower Limit =", lower_limit)
print("Upper Limit =", upper_limit)

plt.boxplot(visitors)
plt.title("Boxplot of Visitors")
plt.ylabel("Visitors")
plt.grid(True)
plt.show()
```

```
Q1 = 29.25
Median (Q2) = 39.0
Q3 = 50.5
IQR = 21.25
Lower Limit = -2.625
Upper Limit = 82.375
```